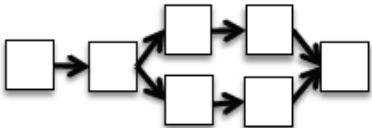
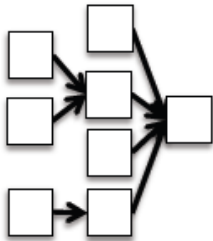
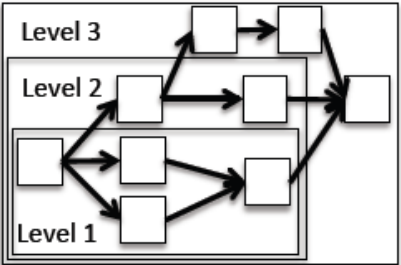
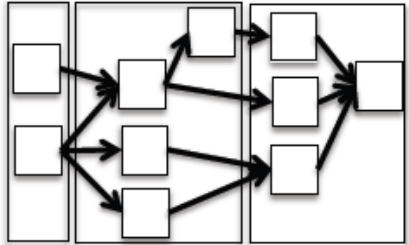


Different Shapes of a Learning Trajectory

 (a) Learning Progression	Definition: A path (or choice of paths) that a student goes through while developing understanding of a particular topic (i.e., the final learning goal) – Must have a starting point and a final point/goal – The final goal is some level of understanding appropriate to the learner's age and experience – Is largely linear with some choices in path
 (b) Pieces of Knowledge	Argument: A particular level of understanding requires different pieces of knowledge. These pieces of knowledge... – may NOT have a particular order, and – may be activated differently depending on the learning context With many more independent pieces of knowledge that, once learned, result in a deeper or more complete understanding (i.e., final learning goal)
 (c) Spiral Curriculum	Hypothesis: Any subject can be taught in some intellectually honest form to any child at any stage of development – Complex topics/concepts can be simplified in various ways for young children – Same topics/concepts revisited multiple times – Revisited in more depth each time – Connections made to previous visit
Everyday Experiences CT Programming  (d) Constructivism	Theory: Constructivism posits that all effective learning builds upon knowledge that learners already have General steps to build a constructionist trajectory: – Start with everyday experiences or knowledge – "Filters" them to identify the aspects that are most relevant to the new learning (i.e., the academic topics/concepts being taught) – Gradually refine everyday knowledge to apply to academic topics/concepts of interest

Examples K-8 computational thinking learning trajectories (Rich et al., 2017 and 2019)

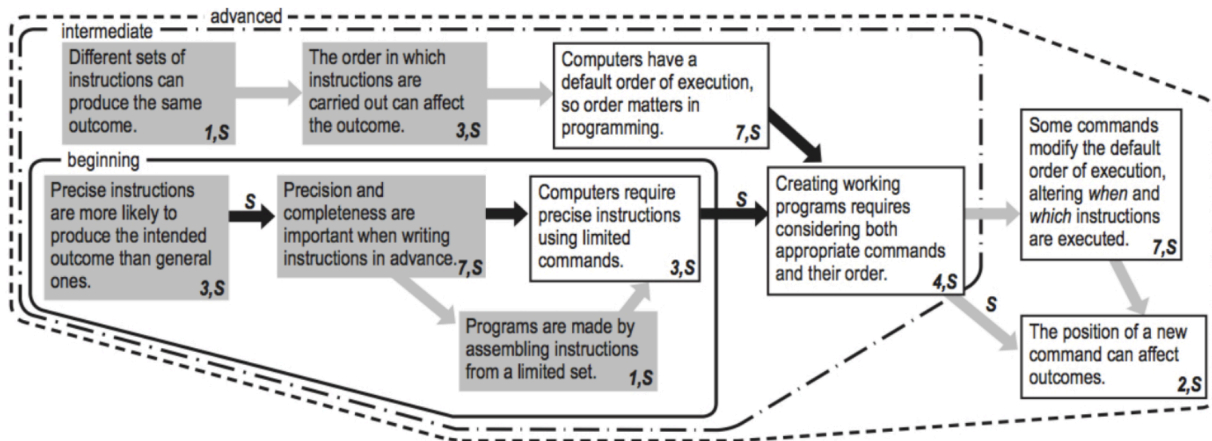


Figure 3: Sequence learning trajectory.

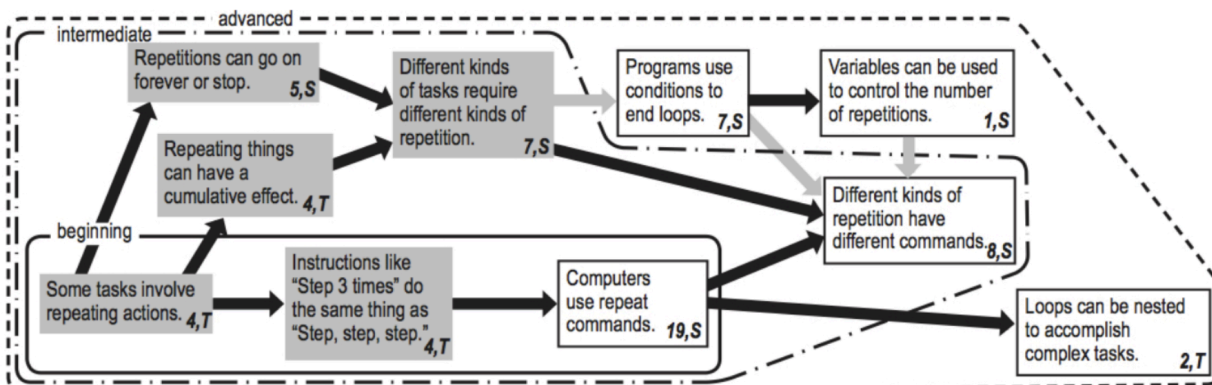


Figure 4: Repetition learning trajectory.

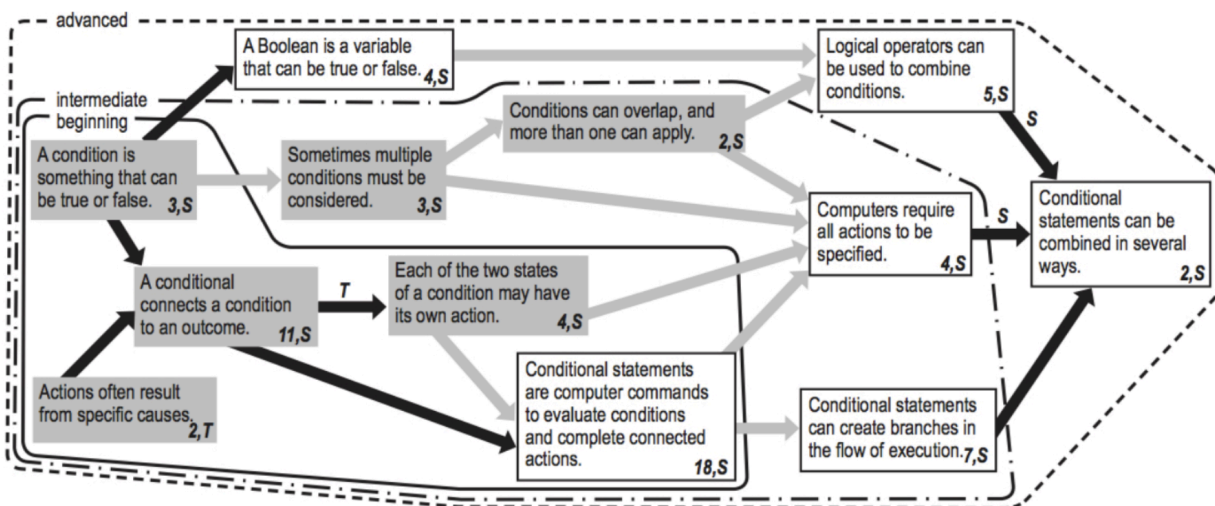


Figure 5: Conditionals learning trajectory.

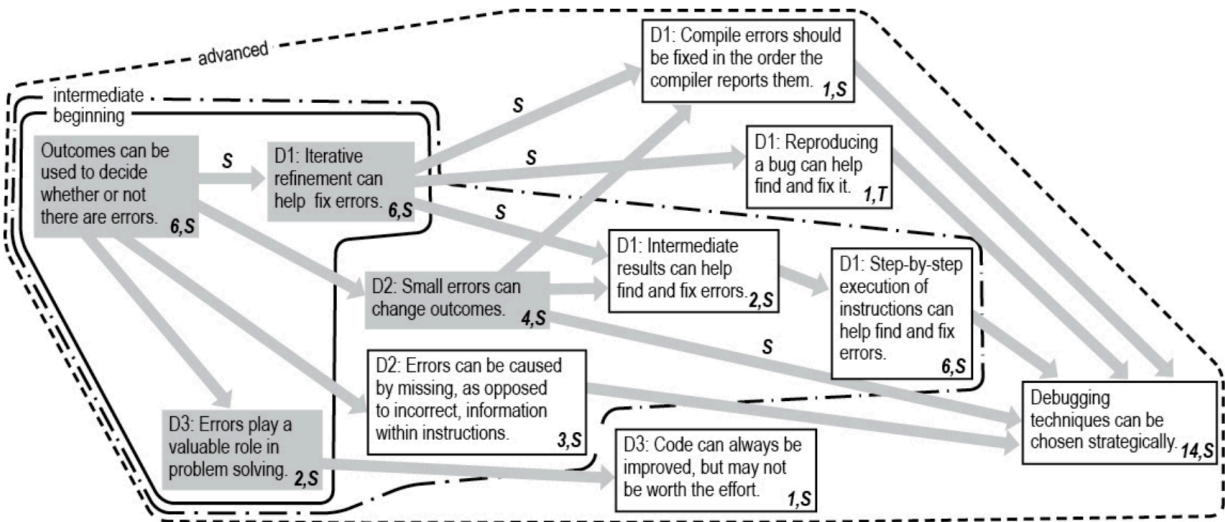


Figure 1: The Debugging Trajectory